

Experiment no - 4

DATE / / 2026

Sl No	Rel ⁿ between P & M	Hydroxide (OH ⁻) in PPM	Carbonate in PPM	Bicarbonate in PPM
	P = 0	0	0	0
1	P = M	0	0	M
2	P = 1/2 M	M	0	0
3	P > 1/2 M	0	2P	0
4	P ≥ 1/2 M	[2P - M]	2[M - P]	0
5	P < 1/2 M	0	2P	[M - 2P]

Observation Table and Calculations

Given: Normality of HCl = 0.02N, V = 10ml
Phenolphthaleine

$$\text{alkalinity 'P' PPM} = \frac{V_1 \times N \times 50 \times 1000}{V}$$

$$\text{methyl orange alkalinity 'M' M} = \frac{V_2 \times N \times 50 \times 1000}{V}$$

Water sample	V ₁ ml	V ₂ ml	Relation between P and M	OH ⁻
Water sample I	4	4.5	P > 1/2 M	OH ⁻
Water sample II	2.5	4	P > 1/2 M	OH ⁻
Water sample III	-	-	-	-

CO_3^{2-}	HCO_3^-
0	0
0	0
CO_3^{2-}	0
0	0
0	0
0	0

Result

Alkalinity Experiment:

Alkalinity water Sample	CO_3^{2-}	HCO_3^-
Alkalinity water 1	850ppm	100ppm
Alkalinity water 2	100ppm	800ppm
Alkalinity water 3	-	-